

Ms Davis
Mrs Greenberg
Mrs Millar

Mr Feng
Ms Lau
Mrs Squires

Name:.....

Teacher's Name:.....



Pymble Ladies' College

**YEAR 11
YEARLY EXAMINATION
2019**

Mathematics Extension 1

**General
Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Erasable pens are **not** to be used
- NESA approved calculators may be used
- Reference sheets are provided

**Total
Marks:
70**

Section I 10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II 60 marks

- Attempt all Questions 11 - 30
- Allow about 1 hours and 45 minutes for this section



NSW Education Standards Authority

2020 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x-h)^2 + (y-k)^2 = r^2$$

Financial Mathematics

$$A = P(1+r)^n$$

Sequences and series

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}, r \neq 1$$

$$S = \frac{a}{1-r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

AutoMarque Answer Sheet ©

Student's name

Yr 11 Extension Mathematics

Hundreds:

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Tens:

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Units:

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- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D

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Section I

Multiple Choice (10 marks)

Use the Multiple Choice Answer Sheet for questions 1 to 10.

1 What is the remainder when $P(x) = x^3 - x^2 + x + 3$ is divided by $(x-1)$?

- (A) 0
- (B) 2
- (C) 3
- (D) 4

2 Given that $\sin 2\alpha = \frac{7}{10}$ and $\sin \alpha = \frac{2}{5}$, what is the value of $\cos \alpha$?

- (A) $-\frac{7}{4}$
- (B) $\frac{7}{8}$
- (C) $\frac{7}{25}$
- (D) $\frac{14}{25}$

3 A function has the parametric equations $x = \frac{2}{t}$ and $y = 2t^2$.

What is the Cartesian equation of this curve?

- (A) $y = \frac{4}{x}$
- (B) $y = \frac{8}{x}$
- (C) $y = \frac{4}{x^2}$
- (D) $y = \frac{8}{x^2}$

4 Given α, β and γ are the roots of $x^3 - 4x - 1 = 0$, what does $\alpha\beta + \alpha\gamma + \beta\gamma$ equal?

(A) -4

(B) -1

(C) 0

(D) 4

5 The polynomial $P(x) = x^3 + 2x + k$ has $(x - 2)$ as a factor.
What is the value of k ?

(A) -12

(B) -10

(C) 10

(D) 12

6 Which of the following is a simplified expression for $\frac{\sin 2x}{1 - \cos 2x}$?

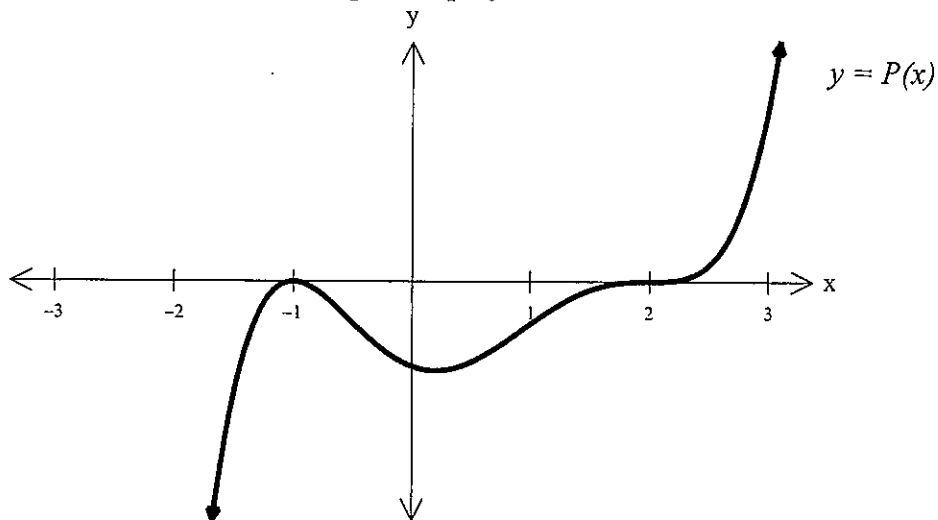
(A) $\sin x$

(B) $\cos x$

(C) $\tan x$

(D) $\cot x$

- 7 The diagram shows the graph of a polynomial.

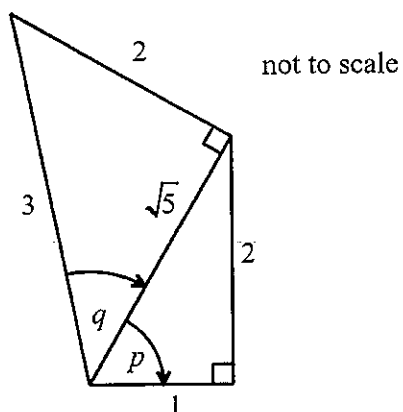


Which of the following could be the equation of this polynomial $P(x)$?

- (A) $P(x) = (x-1)(x+2)$
 - (B) $P(x) = (x-1)^2(x+2)^3$
 - (C) $P(x) = (x+1)^2(x-2)^3$
 - (D) $P(x) = (x+1)(x-2)^3$
- 8 If $\cos^{-1} x + \sin^{-1} x = \frac{\pi}{2}$ and $\cos^{-1} x - \sin^{-1} x = \frac{\pi}{4}$, what is the value of $\cos^{-1} x$?

- (A) $\frac{5\pi}{16}$
- (B) $\frac{3\pi}{8}$
- (C) $\frac{7\pi}{16}$
- (D) $\frac{5\pi}{8}$

- 9 The diagram shows two right-angled triangles with sides and angles given.



What is the value of $\sin(p+q)$?

- (A) $\frac{2}{\sqrt{5}} + \frac{2}{3}$
- (B) $\frac{2}{\sqrt{5}} + \frac{\sqrt{5}}{3}$
- (C) $\frac{2}{3\sqrt{5}} + \frac{2}{3}$
- (D) $\frac{4}{3\sqrt{5}} + \frac{1}{3}$

- 10 What is the value of $\sin^{-1}\left(\sin\frac{5\pi}{7}\right)$?

- (A) $\frac{5\pi}{7}$
- (B) $\frac{2\pi}{7}$
- (C) $\frac{-5\pi}{7}$
- (D) $\frac{-2\pi}{7}$

Section II - Part A (16 marks)

Name.....

Teacher: PL MM JS BD NF DG

11 Find the exact value of $\sin 75^\circ \cos 15^\circ$.

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12 If $t = \tan A$, then find $\sin 2A + \cos 2A$ as a single fraction in terms of t .

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13 Find the exact value of $\sin 15^\circ$.

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14 Solve $\sin \theta = \sin 2\theta$ for $0 \leq \theta \leq 2\pi$.

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15 A curve is represented by the parametric equations $x = 2 \cos \theta$ and $y = 2 \sin \theta$.
Find the Cartesian equation of this curve and describe it geometrically in words.

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Section II - Part B (15 marks)

Name.....

Teacher: PL MM JS BD NF DG

- 18 Two of the roots of $x^3 - px^2 - qx + 30 = 0$ are 3 and 5. Find the value of p and q . 3

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- 19 If $P(x) = x^3 - x^2 - 8x + 12$, show that $P(x)$ has a zero of multiplicity 2. 2

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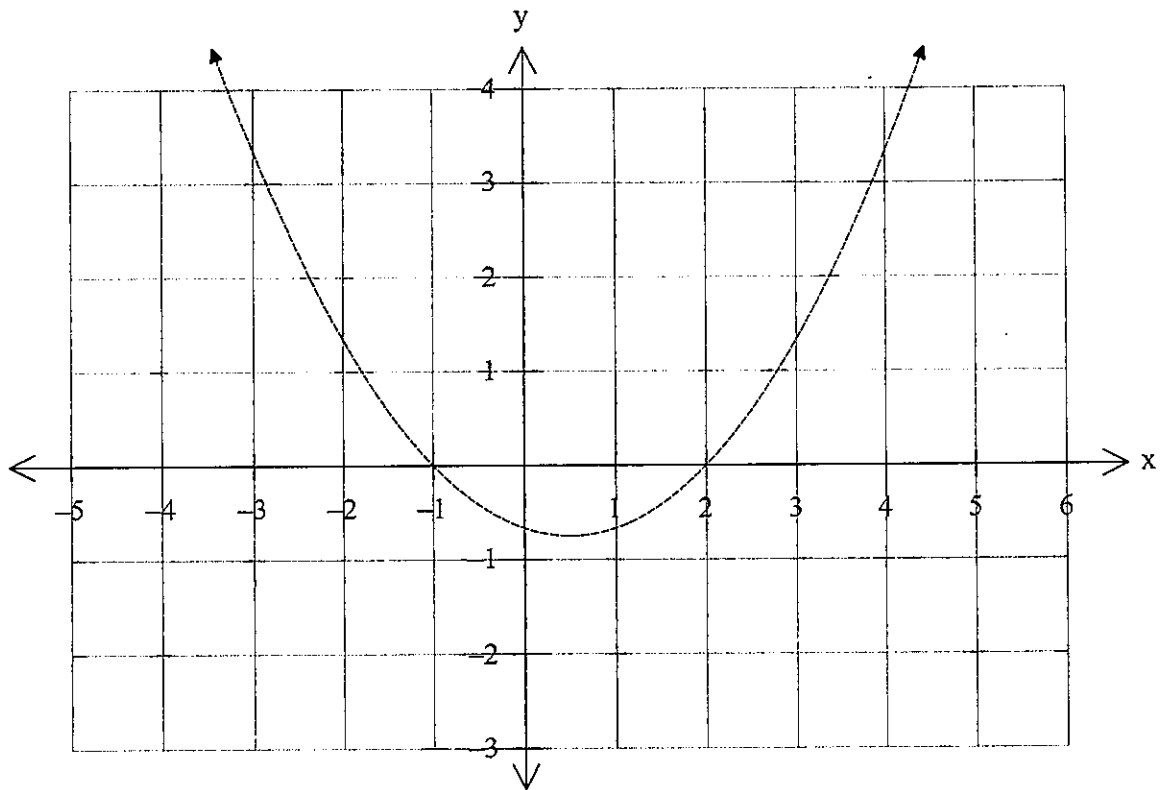
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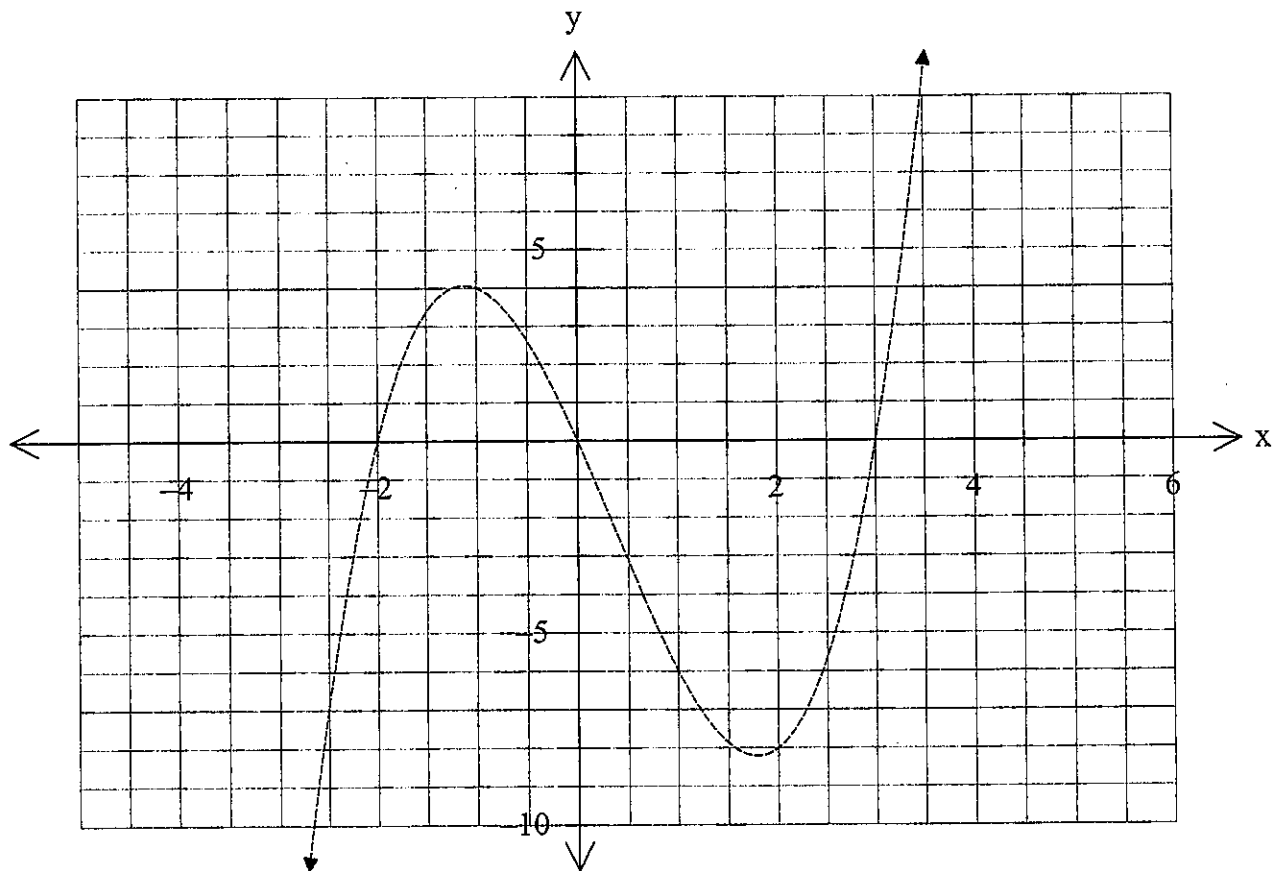
- 20 (a) The diagram shows the graph of $y = f(x)$. Sketch the graph of $y = \sqrt{f(x)}$.

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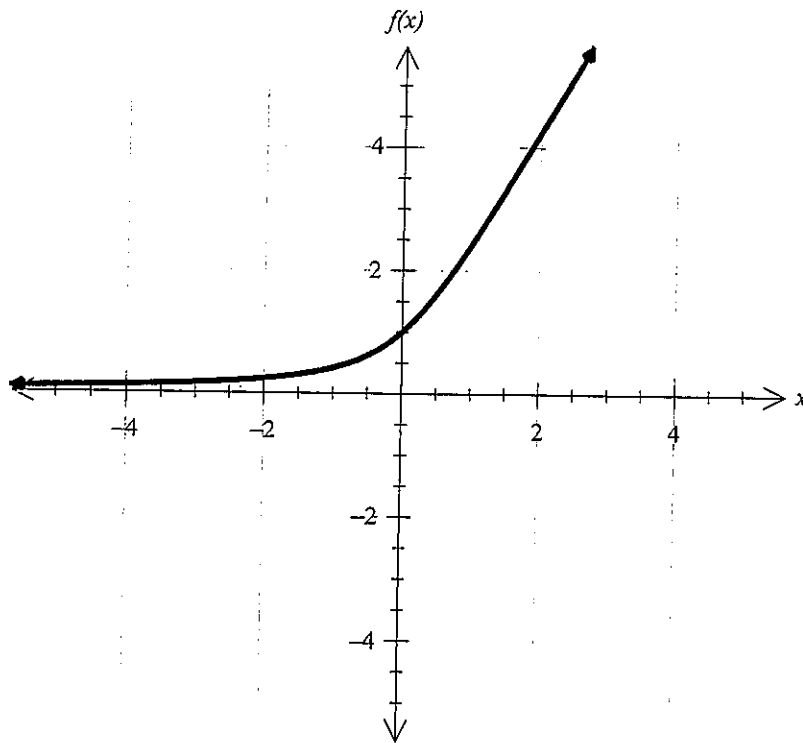


- (b) The diagram shows the graph of $y = f(x)$. Sketch the graph of $y = f(|x|)$.

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23 Consider the function $f(x) = x + \sqrt{x^2 + 1}$.



(a) On the graph above, draw the inverse function of $f(x)$. 2

(b) Show that $f^{-1}(x) = \frac{1}{2}\left(x - \frac{1}{x}\right)$ 3

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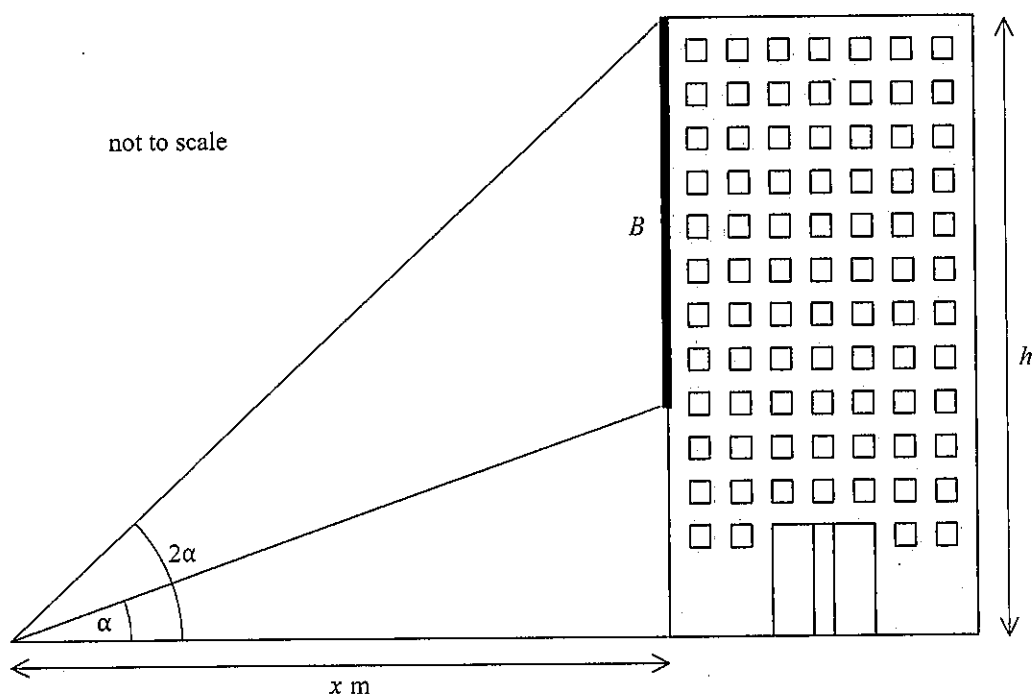
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- 24 Find the exact value of $\cos\left(\tan^{-1}\left(\frac{-1}{3}\right)\right)$.

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- 25 A large advertising banner (B) is hung from the top of a building. The angle of elevation from an observer x metres from the base of the building to the top of the banner is 2α , while the angle of elevation to the bottom of the banner is α .



- (a) Show that the height, h , of the building is $h = x \tan 2\alpha$.

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27 A Cartesian equation can be defined by the parametric equations $x = \frac{t}{t+1}$ and $y = \frac{t}{t-1}$.

(a) Show that $x = \frac{t}{t+1}$ can be rewritten as $t = \frac{x}{1-x}$.

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(b) Hence, find the Cartesian equation defined by the parametric equations $x = \frac{t}{t+1}$ and $y = \frac{t}{t-1}$.

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28 Find the inverse function of $y = \log_2(x-3)$.

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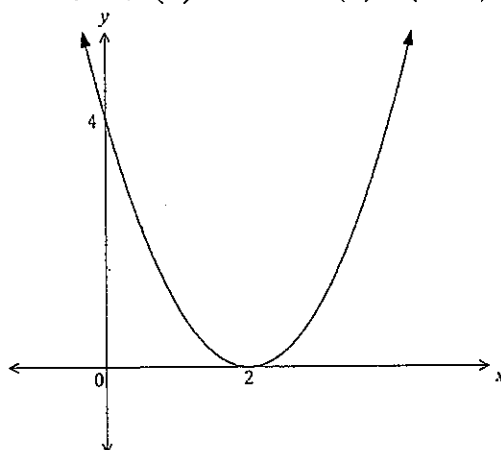
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29 The graph shows the parabola $y = f(x)$, given $f(x) = (x-2)^2$.



(a) Explain why an inverse function $y = f^{-1}(x)$ exists when $x \geq 2$.

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(b) State the domain and range of this inverse function.

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(c) At what point will $f(x) = f^{-1}(x)$?

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End of Paper

Extra writing space

If you use this space, clearly indicate which question you are answering.

Question 25 continued

- (b) Using $t = \tan \alpha$, or otherwise, show that the height, B , of the banner is $B = x \tan \alpha \sec 2\alpha$ metres.

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Section II Part C question 26 is on back page of this booklet.

26 Given $f(x) = x^3 - 1$, $g(x) = 3x + 1$ and $h(x) = 4x - 5$,

(a) show that $g(f(x)) + xh(x) = 3x^3 + 4x^2 - 5x - 2$,

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(b) solve $g(f(x)) + xh(x) = 0$.

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21 Solve $x^2(x-1)(5-x) < 0$.

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22 If $f(x) = x^2 - x - 6$, sketch both $y = f(x)$ and $y = \frac{1}{f(x)}$, on the same number plane. 3

Show all key features and clearly label both graphs.

Extra writing space

If you use this space, clearly indicate which question you are answering.

16 Find the values of x which satisfy $\frac{1}{x+1} \geq 3$.

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17 Given $f(x) = 2x + 3$ and $g(x) = |x - 1|$, find the value(s) of x such that $f(x) < g(x)$. 3

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Extra writing space

If you use this space, clearly indicate which question you are answering.

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

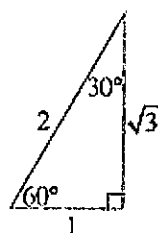
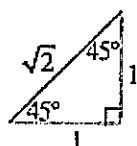
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

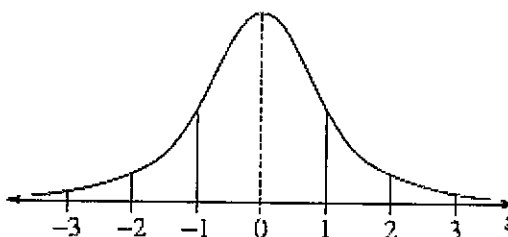
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= {}^nC_x p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$= \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Solutions

Section I

Multiple Choice (10 marks)

Use the Multiple Choice Answer Sheet for questions 1 to 10.

- 1 What is the remainder when $P(x) = x^3 - x^2 + x + 3$ is divided by $(x-1)$?

(A) 0

(B) 2

(C) 3

(D) 4

$$P(1) = (1)^3 - (1)^2 + (1) + 3 \\ = 4$$

- 2 Given that $\sin 2\alpha = \frac{7}{10}$ and $\sin \alpha = \frac{2}{5}$, what is the value of $\cos \alpha$?

(A) $-\frac{7}{4}$

(B) $\frac{7}{8}$

(C) $\frac{7}{25}$

(D) $\frac{14}{25}$

$$\sin 2\alpha = \frac{7}{10} \\ \therefore 2 \sin \alpha \cos \alpha = \frac{7}{10}$$

$$2 \left(\frac{2}{5} \right) \cos \alpha = \frac{7}{10}$$

$$\frac{4}{5} \cos \alpha = \frac{7}{10}$$

$$\cos \alpha = \frac{7}{10} \times \frac{5}{4}$$

$$\cos \alpha = \frac{7}{8}$$

- 3 A function has the parametric equations $x = \frac{2}{t}$ and $y = 2t^2$.

What is the Cartesian equation of this curve?

(A) $y = \frac{4}{x}$

(B) $y = \frac{8}{x}$

(C) $y = \frac{4}{x^2}$

(D) $y = \frac{8}{x^2}$

$$x = \frac{2}{t}$$

$$t = \frac{2}{x}$$

$$y = 2t^2$$

$$y = 2 \left(\frac{2}{x} \right)^2$$

$$\therefore y = \frac{8}{x^2}$$

- 4 Given α, β and γ are the roots of $x^3 - 4x - 1 = 0$, what does $\alpha\beta + \alpha\gamma + \beta\gamma$ equal?

(A) -4

(B) -1

(C) 0

(D) 4

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$= \frac{-4}{1}$$

$$= -4$$

- 5 The polynomial $P(x) = x^3 + 2x + k$ has $(x-2)$ as a factor.

What is the value of k ?

(A) -12

(B) -10

(C) 10

(D) 12

$$P(2) = 0$$

$$0 = (2)^3 + 2(2) + k$$

$$0 = 12 + k$$

$$k = -12$$

- 6 Which of the following is a simplified expression for $\frac{\sin 2x}{1 - \cos 2x}$?

(A) $\sin x$

(B) $\cos x$

(C) $\tan x$

(D) $\cot x$

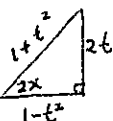
$$= \frac{2 \sin x \cos x}{1 - (1 - 2 \sin^2 x)}$$

$$= \frac{2 \sin x \cos x}{2 \sin^2 x}$$

$$= \frac{\cos x}{\sin x}$$

$$= \cot x$$

OR
Let $t = \tan x$



$$\therefore \frac{2t}{1+t^2} \times \frac{1+t}{1-t^2}$$

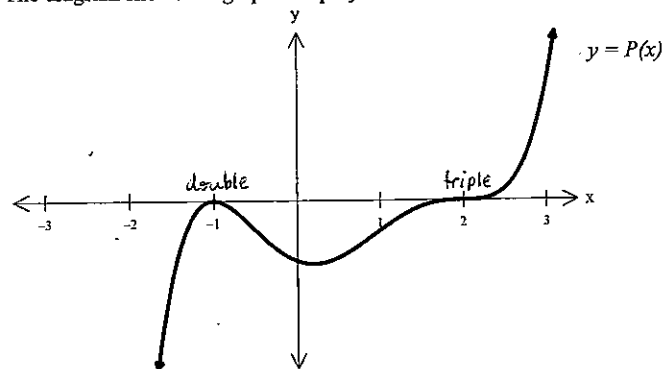
$$= \frac{2t}{1+t^2-1+t^2}$$

$$= \frac{2t}{2t^2}$$

$$= \frac{1}{t}$$

$$= \cot x$$

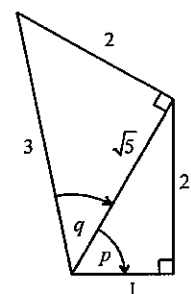
- 7 The diagram shows the graph of a polynomial.



Which of the following could be the equation of this polynomial $P(x)$?

- (A) $P(x) = (x-1)(x+2)$
 (B) $P(x) = (x-1)^2(x+2)^3$
 (C) $P(x) = (x+1)^2(x-2)^3$
 (D) $P(x) = (x+1)(x-2)^3$

- 9 The diagram shows two right-angled triangles with sides and angles given.



not to scale

$$\sin p = \frac{2}{\sqrt{5}} \quad \cos p = \frac{1}{\sqrt{5}}$$

$$\sin q = \frac{2}{3} \quad \cos q = \frac{\sqrt{5}}{3}$$

$$\begin{aligned} \sin(p+q) &= \sin p \cos q + \cos p \sin q \\ &= \frac{2}{\sqrt{5}} \cdot \frac{\sqrt{5}}{3} + \frac{1}{\sqrt{5}} \cdot \frac{2}{3} \\ &= \frac{2}{3} + \frac{2}{3\sqrt{5}} \end{aligned}$$

What is the value of $\sin(p+q)$?

- (A) $\frac{2}{\sqrt{5}} + \frac{2}{3}$
 (B) $\frac{2}{\sqrt{5}} + \frac{\sqrt{5}}{3}$
 (C) $\frac{2}{3\sqrt{5}} + \frac{2}{3}$
 (D) $\frac{4}{3\sqrt{5}} + \frac{1}{3}$

- 8 If $\cos^{-1} x + \sin^{-1} x = \frac{\pi}{2}$ and $\cos^{-1} x - \sin^{-1} x = \frac{\pi}{4}$, what is the value of $\cos^{-1} x$?

- (A) $\frac{5\pi}{16}$
 (B) $\frac{3\pi}{8}$
 (C) $\frac{7\pi}{16}$
 (D) $\frac{5\pi}{8}$

$$\begin{aligned} A + B \\ \cos^{-1} x + \sin^{-1} x &= \frac{\pi}{2} \quad - [1] \end{aligned}$$

$$\begin{aligned} A - B \\ \cos^{-1} x - \sin^{-1} x &= \frac{\pi}{4} \quad - [2] \end{aligned}$$

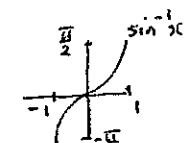
$$\begin{aligned} [1] + [2]: \quad 2 \cos^{-1} x &= \frac{3\pi}{4} \\ \cos^{-1} x &= \frac{3\pi}{8} \end{aligned}$$

- 10 What is the value of $\sin^{-1}\left(\sin \frac{5\pi}{7}\right)$?

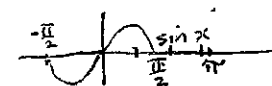
- (A) $\frac{5\pi}{7}$
 (B) $\frac{2\pi}{7}$
 (C) $\frac{-5\pi}{7}$
 (D) $\frac{-2\pi}{7}$

$$\begin{aligned} A &= \sin^{-1}\left(\sin \frac{5\pi}{7}\right) \\ \sin A &= \sin \frac{5\pi}{7} \\ \sin A &= \sin\left(\pi - \frac{2\pi}{7}\right) \\ \sin A &= \sin \frac{2\pi}{7} \end{aligned}$$

$$-1 \leq x \leq 1$$



$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$



11 Find the exact value of $\sin 75^\circ \cos 15^\circ$.

1

$$\sin 75^\circ \cos 15^\circ = \frac{1}{2} [\sin 90^\circ + \sin 60^\circ]$$

* students who used a product of compound results were less successful.

eg $\sin(30^\circ + 45^\circ) \cos(45^\circ - 30^\circ) = \frac{2 + \sqrt{3}}{4}$ due to alg. errors.

$$= \frac{1}{2} \left[1 + \frac{\sqrt{3}}{2} \right]$$

$$= \frac{1}{2} \cdot \frac{(2 + \sqrt{3})}{2}$$

- 11 RW
- students were expected to write as a sum & substitute correctly
 - ISE

12 If $t = \tan A$, then find $\sin 2A + \cos 2A$ as a single fraction in terms of t .

2

$$\sin 2A + \cos 2A, \quad t = \tan A$$

$$= \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}$$

$$= \frac{2t+1-t^2}{1+t^2}$$

$$= \frac{-t^2 + 2t + 1}{1+t^2}$$

2 correct soln • ISE

1 identify half angle result & substitute

- final answer had to be a single simplified fraction in terms of t .
- students who did not use $t = \tan A$ were given an eq in some cases
- no errors in algebra
- relationship easily recalled

13 Find the exact value of $\sin 15^\circ$.

$$\sin 15^\circ = \sin(60^\circ - 45^\circ)$$

$$= \sin 60^\circ \cos 45^\circ - \cos 60^\circ \sin 45^\circ \quad \text{OR}$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6}-\sqrt{2}}{4}$$

$$\sin 15^\circ = \sin(45^\circ - 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6}-\sqrt{2}}{4}$$

3 correct solution • must be rationalised

2 substitute correctly

1 expanded using compound angle result

14 Solve $\sin \theta = \sin 2\theta$ for $0 \leq \theta \leq 2\pi$.

2

$$\sin \theta = 2 \sin \theta \cos \theta$$

$$0 = 2 \sin \theta \cos \theta - \sin \theta$$

$$0 = \sin \theta (2 \cos \theta - 1)$$

$$\therefore \sin \theta = 0 \quad \text{or} \quad \cos \theta = \frac{1}{2}$$

$$\theta = 0, \pi, 2\pi$$

$$\theta = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\text{Sol'n: } \theta = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

2 correct solutions (5)

1 one correct set of solutions or one error made

* many students not using radians but converting from degrees → errors made in conversion

* many students divided by $\sin \theta$ and lost a set of solutions.

15 A curve is represented by the parametric equations $x = 2 \cos \theta$ and $y = 2 \sin \theta$. Find the Cartesian equation of this curve and describe it geometrically in words.

2

$$x = 2 \cos \theta$$

$$\cos \theta = \frac{x}{2}$$

$$y = 2 \sin \theta$$

$$\sin \theta = \frac{y}{2}$$

$$\cos^2 \theta + \sin^2 \theta = 1 \quad (\text{Pythag. Id})$$

$$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$

$$\frac{x^2}{4} + \frac{y^2}{4} = 1$$

$$x^2 + y^2 = 4$$

Circle centre (0,0), radius = 2 units

2 correct solution

• students were expected to describe shape & features correctly in words

1 equation in cartesian form

* many students factorised incorrectly $-t^2 + 2t + 1 \Rightarrow (t+1)^2$ or $(t-1)^2$

no! no! no!

* many students did not achieve 3 because they did not rationalise

16 Find the values of x which satisfy $\frac{1}{x+1} \geq 3$.

3

Extra writing space

If you use this space, clearly indicate which question you are answering.

$$\frac{(x+1)^2}{x+1} \geq 3(x+1)^2 \quad ; x \neq -1$$

$$x+1 \geq 3(x+1)^2 \quad ; x \neq -1$$

$$0 \geq 3(x+1)^2 - (x+1) \quad ; x \neq -1$$

$$0 \geq (x+1)[3(x+1) - 1] \quad ; x \neq -1$$

$$0 \geq (x+1)(3x+2) \quad ; x \neq -1$$



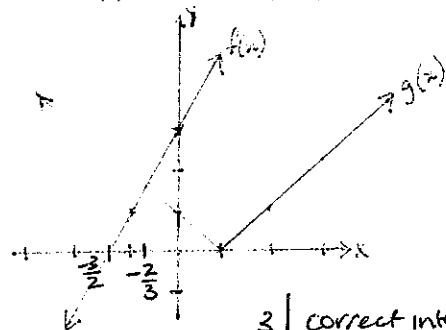
$$\text{Sol'n: } -1 < x \leq -\frac{2}{3}$$

3	correct solution solution needed to exclude $x = -1$
2	simplified quadratic inequality
1	multiply by $(x+1)^2$ both sides

⊗ one student sketched a hyperbola and solved, after finding the point of intersection.

⊗ students who did not sketch the quadratic were often unsuccessful.

17 Given $f(x) = 2x+3$ and $g(x) = |x-1|$, find the value(s) of x such that $f(x) < g(x)$. 3



$$f(x) < g(x)$$

$$2x+3 < -x+1$$

$$3x < -2$$

$$x < -\frac{2}{3}$$

$$\text{Sol'n } x < -\frac{2}{3}$$

3	correct interpretation of solution to inequalities.
2	inequalities solved correctly.
1	inequality correctly set up with abs value.

⊗ working was difficult to read with many students leaving two regions without reasoning.

⊗ students are encouraged to sketch neat diagrams that are labelled. This reduced the amount of work required to solve the problem.

⊗ many students were unable to determine why $x < -4$ was not the correct solution.

⊗ lots of errors with manipulating negatives around inequalities.

If you use this space, clearly indicate which question you are answering.

$$\alpha = 3, \beta = 5, \gamma = 8$$

$$\alpha + \beta + \gamma = \frac{-b}{a}$$

$$8 + 5 = p \quad \text{--- (1)}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$15 + 3\gamma + 5\gamma = -q$$

$$15 + 8\gamma = -q \quad \text{--- (2)}$$

$$\alpha\beta\gamma = \frac{-d}{a}$$

$$15\gamma = -30$$

$$\gamma = -2$$

Subst. $\gamma = -2$ into (1): $p = 8 + (-2)$

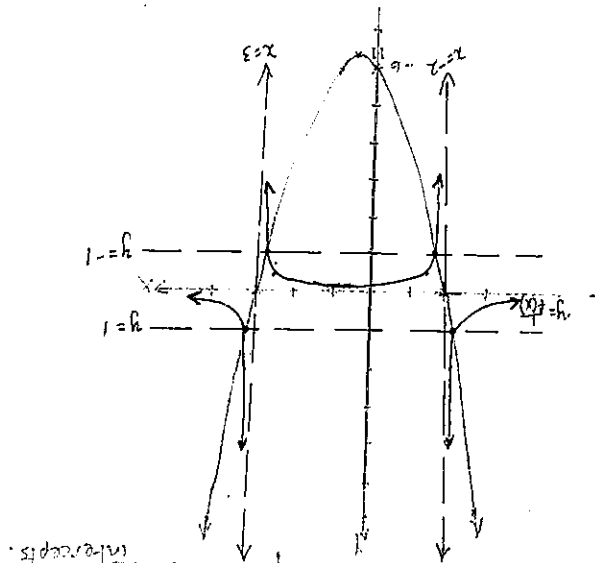
$$p = 6$$

Subst. $\gamma = -2$ into (2): $-q = 15 + 8(-2)$

$$-q = -1$$

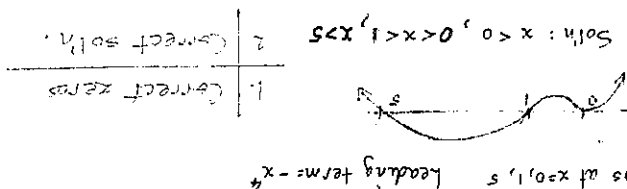
$$q = 1$$

12



1 Vertical asymptotes.
2 Correct shape.
3 Correct $y = 1, y = -1$ intercepts.

22 If $f(x) = x^3 - x^2 - 6x + 6$, sketch both $y = f(x)$ and $y = \frac{1}{f(x)}$ on the same number plane.



Zeros at $x = 0, 1, 3$ leading term $= -x^3$

21 Solve $x^2(x-1)(x-3) < 0$.

18 Two of the roots of $x^3 - px^2 - qx + 30 = 0$ are 3 and 5. Find the value of p and q .

$$P(3) = 0$$

$$(3)^3 - p(3)^2 - q(3) + 30 = 0$$

$$27 - 9p - 3q + 30 = 0$$

$$9p + 3q = 57$$

$$3p + q = 19 \quad \text{--- (1)}$$

$$P(5) = 0$$

$$(5)^3 - p(5)^2 - q(5) + 30 = 0$$

$$125 - 25p - 5q + 30 = 0$$

$$25p + 5q = 155$$

$$5p + q = 31 \quad \text{--- (2)}$$

$$[2] - [1]: 2p = 12$$

$$p = 6$$

Subst. $p = 6$ into (1):

$$3(6) + q = 19$$

$$q = 1$$

$$\text{Sol'n: } \begin{cases} p = 6 \\ q = 1 \end{cases}$$

Either [1] or [2]

2 Solve for either p or q.
3 Correct sol'n

19 If $P(x) = x^3 - x^2 - 8x + 12$, show that $P(x)$ has a zero of multiplicity 2.

$$P(x) = x^3 - x^2 - 8x + 12$$

$$P'(x) = 3x^2 - 2x - 8$$

$$0 = 3x^2 - 2x - 8$$

$$= (3x+4)(x-2)$$

$$x = -\frac{4}{3} \text{ or } 2$$

$$P(2) = (2)^3 - (2)^2 - 8(2) + 12 = 0$$

$$P(2) = P'(2) = 0$$

$\therefore$ Zero of multiplicity 2.

1 Find $P(2) = 0$

2 Find $P(2) = P'(2) = 0$

or Find fully factorised form.

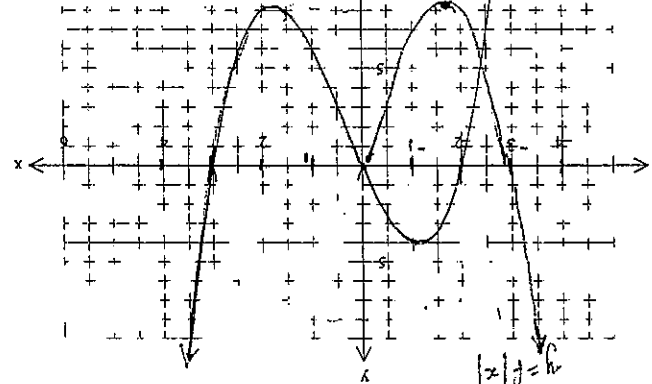
$$\begin{aligned} P(x) &= (x-2)(x^2+x-6) \\ &= (x-2)(x+3)(x-2) \\ &= (x-2)^2(x+3) \end{aligned}$$

$$\begin{aligned} & \frac{x^3+x-6}{x-2} \\ & \frac{x^3-x^2-8x+12}{x^3-2x^2} \\ & \frac{x^2-8x}{x^2-2x} \\ & \frac{-6x+12}{-6x+12} \end{aligned}$$

1 x-intercepts

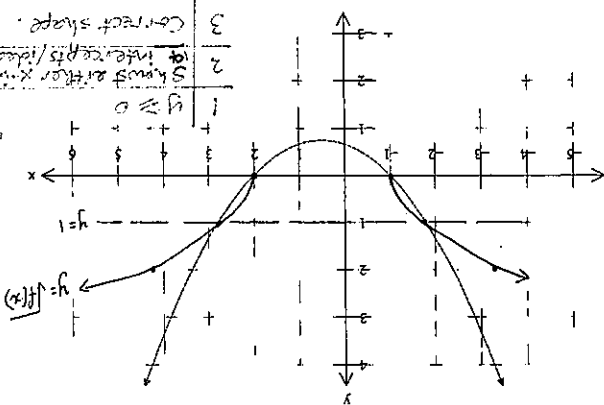
2 Correct point P

3 Correct shape



(b) The diagram shows the graph of $y = f(x)$. Sketch the graph of $y = f(|x|)$.

1 Sketch $y = f(x)$ for $x \geq 0$
2 Sketch $y = f(x)$ for $x < 0$
3 Correct shape.



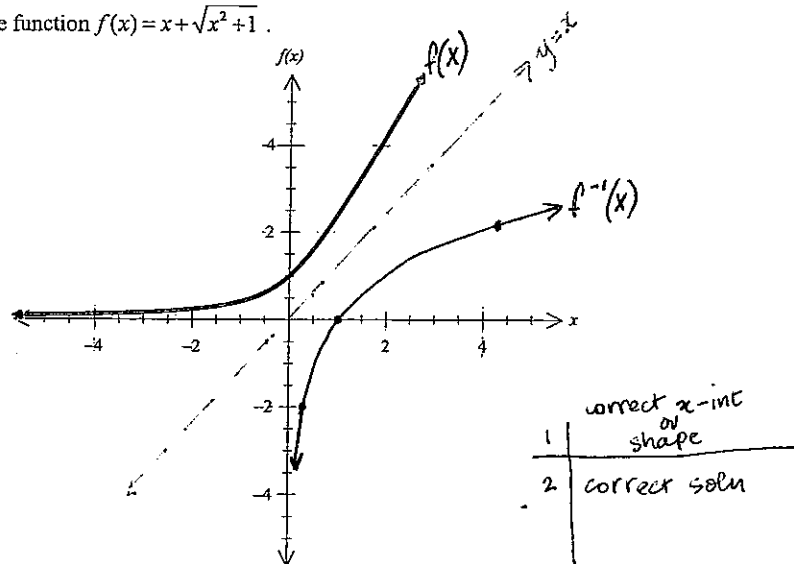
(a) The diagram shows the graph of $y = f(x)$. Sketch the graph of $y = \sqrt{f(x)}$.

Section II - Part C (16 marks)

Name.....

Teacher: PL MM JS BD NF DG

- 23 Consider the function $f(x) = x + \sqrt{x^2 + 1}$.



- (a) On the graph above, draw the inverse function of $f(x)$.

2

- (b) Show that $f^{-1}(x) = \frac{1}{2}(x - \frac{1}{x})$

3

Let $y = x + \sqrt{x^2 + 1}$

f^{-1} : $x = y + \sqrt{y^2 + 1}$

$x - y = \sqrt{y^2 + 1}$

$(x - y)^2 = y^2 + 1$

$x^2 - 2xy + y^2 = y^2 + 1$

$x^2 - 2xy = 1$

$x^2 - 1 = 2xy$

$\frac{x^2 - 1}{2x} = y$; $x \neq 0$

$y = \frac{x^2}{2x} - \frac{1}{2x}$

$y = \frac{x}{2} - \frac{1}{2x}$

$\therefore f^{-1}(x) = \frac{1}{2}(x - \frac{1}{x})$

1	$x = y + \sqrt{y^2 + 1}$
2	works towards $x^2 - 2xy + y^2 = y^2 + 1$
3	$f^{-1}(x) = \frac{x}{2} - \frac{1}{2x}$

13

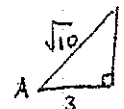
- 24 Find the exact value of $\cos\left(\tan^{-1}\left(\frac{-1}{3}\right)\right)$.

2

$A = \tan^{-1}\left(\frac{-1}{3}\right)$

$\therefore \tan A = \frac{-1}{3}$

$\frac{S}{T} = \frac{A}{C}$



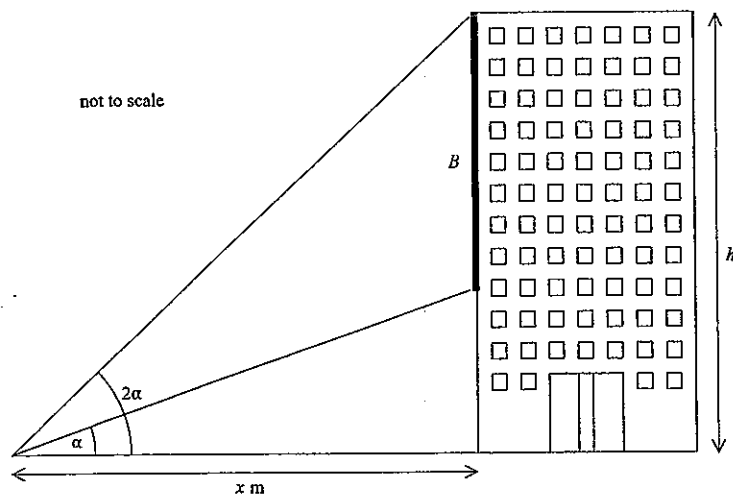
$\tan$ -ve in (Q4)

$\therefore \cos A = \frac{3}{\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}}$

$\cos A = \frac{3\sqrt{10}}{10}$

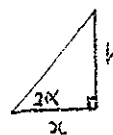
1	$\tan A = -\frac{1}{3}$
2	correct soln $\frac{3}{\sqrt{10}} \approx \frac{3\sqrt{10}}{10}$

- 25 A large advertising banner (B) is hung from the top of a building. The angle of elevation from an observer x metres from the base of the building to the top of the banner is 2α , while the angle of elevation to the bottom of the banner is α .



- (a) Show that the height, h , of the building is $h = x \tan 2\alpha$.

1



$\tan 2\alpha = \frac{h}{x}$

$h = x \cdot \tan 2\alpha$... As req'd.

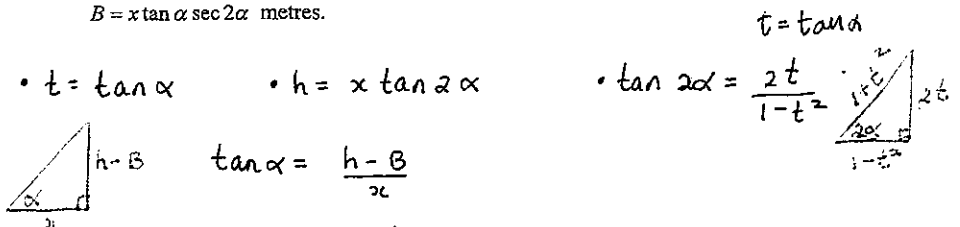
1	correct soln
---	--------------

14

Question 25 continued

- (b) Using $t = \tan \alpha$, or otherwise, show that the height, B , of the banner is
 $B = x \tan \alpha \sec 2\alpha$ metres.

3



$$\tan \alpha = \frac{h-B}{x}$$

$$h-B = x \tan \alpha$$

$$B = h - x \tan \alpha$$

$$= x \tan 2\alpha - x \tan \alpha$$

$$= x \times \frac{2t}{1-t^2} - xt$$

$$= xt \left(\frac{2}{1-t^2} - 1 \right)$$

$$= xt \left(\frac{2 - (1-t^2)}{1-t^2} \right)$$

$$= xt \left(\frac{1+t^2}{1-t^2} \right)$$

$$= xt \left(\frac{1}{\cos 2\alpha} \right)$$

$$= x \cdot \tan \alpha \cdot \sec 2\alpha \quad \dots \text{As req'd.}$$

1	$\tan 2\alpha = \frac{2t}{1-t^2}$ or $\tan \alpha = \frac{h-b}{x}$
2	$B = h - x \cdot \tan \alpha$ $B = x \cdot \tan 2\alpha - x \cdot \tan \alpha$ attempts to replace $\tan 2\alpha$ and $\tan \alpha$ with t -formulae
3	correct conclusion

- 26 Given $f(x) = x^3 - 1$, $g(x) = 3x + 1$ and $h(x) = 4x - 5$,

- (a) show that $g(f(x)) + xh(x) = 3x^3 + 4x^2 - 5x - 2$,

2

$$\begin{aligned} g(f(x)) &= g(x^3 - 1) \\ &= 3(x^3 - 1) + 1 \\ &= 3x^3 - 2 \end{aligned}$$

$$\begin{aligned} g(f(x)) + x \cdot h(x) &= 3x^3 - 2 + x(4x - 5) \\ &= 3x^3 - 2 + 4x^2 - 5x \\ &= 3x^3 + 4x^2 - 5x - 2 \end{aligned}$$

1	$g(f(x))$ $= 3(x^3 - 1) + 1$
2	correct soln

- (b) solve $g(f(x)) + xh(x) = 0$.

$$3x^3 + 4x^2 - 5x - 2 = 0$$

$$P(1) = 3(1)^3 + 4(1)^2 - 5(1) - 2$$

$$= 3 + 4 - 5 - 2$$

$$= 0$$

$\therefore (x-1)$ is a factor.

$$\begin{array}{r} 3x^2 + 7x + 2 \\ x-1 \overline{) 3x^3 + 4x^2 - 5x - 2} \end{array}$$

$$\underline{3x^3 - 3x^2}$$

$$7x^2 - 5x$$

$$\underline{7x^2 - 7x}$$

$$2x - 2$$

$$\underline{2x - 2}$$

$$\text{i.e. } 3x^3 + 4x^2 - 5x - 2 = 0$$

$$(x-1)(3x^2 + 7x + 2) = 0$$

$$(x-1)(3x+1)(x+2) = 0$$

$$x = -2, -\frac{1}{3}, 1$$

$$\begin{array}{c} 3x \\ \times \\ x \end{array}$$

1	$P(1) \neq 0$ or $P(-2) = 0$
2	Finds other factor by long ÷ or roots are $-2, 1, -\frac{1}{3}$ $\therefore x-1 = \frac{x}{3}$ $x = -\frac{1}{3}$
3	soln: $x = -2$ or $-\frac{1}{3}$ or 1

- 27 A Cartesian equation can be defined by the parametric equations $x = \frac{t}{t+1}$ and $y = \frac{t}{t-1}$.

- (a) Show that $x = \frac{t}{t+1}$ can be rewritten as $t = \frac{x}{1-x}$.

2

$$\begin{aligned} x &= \frac{t}{t+1} \\ x(t+1) &= t \\ xt + x &= t \\ x &= t - xt \\ x &= t(1-x) \\ \therefore t &= \frac{x}{1-x} \quad \dots \text{As req'd} \end{aligned}$$

- (b) Hence, find the Cartesian equation defined by the parametric equations $x = \frac{t}{t+1}$ and $y = \frac{t}{t-1}$.

2

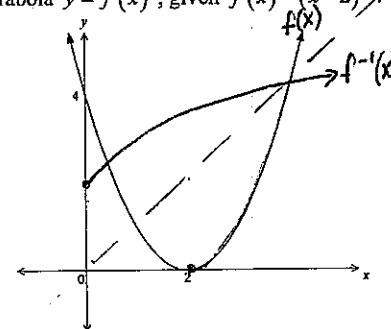
$$\begin{aligned} \text{Subst. } t &= \frac{x}{1-x} \text{ into } y = \frac{t}{t-1} \\ &= \frac{\frac{x}{1-x}}{\frac{x}{1-x} - 1} \\ &= \frac{\frac{x}{1-x}}{\frac{x - (1-x)}{1-x}} = \frac{x}{1-x} \cdot \frac{1-x}{x - (1-x)} \\ &= \frac{x}{2x-1} \end{aligned}$$

- 28 Find the inverse function of $y = \log_2(x-3)$.

$$\begin{aligned} \text{Inverse fn: } x &= \log_2(y-3) \\ y-3 &= 2^x \\ y &= 2^x + 3 \end{aligned}$$

(2)

- 29 The graph shows the parabola $y = f(x)$, given $f(x) = (x-2)^2$.



- (a) Explain why an inverse function $y = f^{-1}(x)$ exists when $x \geq 2$.

1

For $x \geq 2$, $f(x)$ is monotonically increasing,
i.e. It passes the horizontal line test.
 $\therefore$ When $x \geq 2$, $y = f^{-1}(x)$ exists.

(1)

- (b) State the domain and range of this inverse function.

2

$$\begin{aligned} f^{-1}(x): \quad D \quad x: x \geq 2, x \in \mathbb{R} \\ R \quad y: y \geq 2, y \in \mathbb{R} \end{aligned}$$

(2)

- (c) At what point will $f(x) = f^{-1}(x)$?

1

Meet on line of reflection $y = x$:

$$\begin{aligned} y &= (x-2)^2 \\ \text{i.e. } x &= (x-2)^2 \\ x &= x^2 - 4x + 4 \\ 0 &= x^2 - 5x + 4 \\ 0 &= (x-4)(x-1) \\ x &= 1 \text{ or } 4 \end{aligned}$$

(1)

For $x \geq 2$, Point $= (4, 4)$

30 Prove that $\frac{\cos x - \cos(x+2\theta)}{2\sin\theta} = \sin(x+\theta)$.

3

$$\begin{aligned}
 \text{LHS} &= \frac{\cos x - \cos(x+2\theta)}{2\sin\theta} \\
 &= \frac{\cos x - (\cos x \cos 2\theta + \sin x \sin 2\theta)}{2\sin\theta} \\
 &= \frac{\cos x - \cos x (1 - 2\sin^2\theta) + \sin x (2\sin\theta \cos\theta)}{2\sin\theta} \\
 &= \frac{\cos x - \cos x + 2\cos x \sin^2\theta + 2\sin x \sin\theta \cos\theta}{2\sin\theta} \\
 &= \frac{2\cos x \sin^2\theta + 2\sin x \sin\theta \cos\theta}{2\sin\theta} \\
 &= \frac{2\sin\theta (\cos x \sin\theta + \sin x \cos\theta)}{2\sin\theta} \\
 &= \sin x \cos\theta + \cos x \sin\theta \\
 &= \sin(x+\theta) \\
 &= \text{RHS} \quad \dots \text{As req'd}
 \end{aligned}$$

3

* Do not cross out when simplifying.

Extra writing space

If you use this space, clearly indicate which question you are answering.

End of Paper